From Lean proofs to certified derivations in the ontological embedding

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1 Introduction

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Setting

The companion findings on this site give v4 [6] a machine-checked semantics: a deep proof calculus, sound and complete over $\mathcal{L}$-general models. Its derivations were written by hand. The shallow embedding's Lean proofs are richer than higher-order logic, which is how a nominal can hide in a proof that v4's detector passes. This finding builds a Lean command that reads a proof and produces the certified derivation itself.

Result

- **The translator.** For a theorem of a shallow v4 module whose proof stays within higher-order logic (introduction and elimination of the connectives and quantifiers, classical logic, `Eq.subst`, `congrArg`, `congrFun`, `funext`, `propext`, beta and eta, unfolding of the module's definitions), `#deep_translate X as N` reads the statement and the proof term and builds a derivation in the certified calculus, kernel-checked, whose conclusion is the deep image of the statement, matched by `rfl`. It also reads off the module's axioms the proof uses as an $\mathcal{L}$-theory T, and yields `N.valid`: the statement holds at every world of every v4 $\mathcal{L}$-general model of T with nonempty individuals. A proof outside the fragment, or one violating the whole-term condition, is rejected with the first offending step named. The rejections are tests the build enforces: v4's own hybrid proof of `L_R`, the detector-passing proof with a hidden nominal, and a minimal `congrFun` probe.

- **Applied to v4's proofs.** L under the rigid reading (from the detector-certified proof of `L_R`), unrestricted Ax1Gen from Ax1GenTwo in variants 2 and $3'$, modal collapse, necessary and possible existence of a God-

like being, and the essence theorems (`EssMemI`, `UniqueEss1`, `UniqueEss2`, `Ess_iff_Ess2`) in variant 2, $3'$ and in v4's S4 module of variant 3, each now as a semantic theorem over $\mathcal{L}$-general models.

- **The frame condition for Th3.** For Th3 ($\Diamond\exists^E G \to \Box\exists^E G$) from Ax2a, Ax2b and Ax3 in variant 2, translation gives validity on symmetric and on Euclidean frames. Reflexive and transitive frames do not suffice, nor does any set of frame formulas true on a fixed five-edge frame. The exact condition: Th3 holds everywhere in every $\mathcal{L}$-general model on (W, R) iff no world outside an R-closed set B of all-dead or all-serial worlds sees both a member and a non-member of B.

Why it matters

A detector certifies a proof only as far as the detector reads it correctly. The translator replaces that trust with a kernel-checked derivation and the calculus's soundness theorem, and it turns v4's Lean proofs into statements about v4's semantics without rewriting them by hand. The Th3 result shows what the semantics adds: an exact frame condition, where the shallow embedding only gives sufficient ones.

Proved vs checked

Everything is Lean 4 (core library, axioms propext, Classical.choice and Quot.sound only) and is kernel-checked in the library build. The run's skeptic verified the translator's first three results (L under the rigid reading; the equivalence in variants 2 and $3'$); the remaining thirteen claims carry the status *analogy*, because the run ended before its skeptic checked them. We rebuilt six of them outside the run and checked their axioms: the symmetric-frame result, the reflexive–transitive and five-edge counterexamples, the general countermodel construction, the sufficiency direction, and the exact condition.

Novelty statement

This result extends:

- C. Benzmüller, "Proofs Without Nominals ...", arXiv 2609.36279 v4 (https://arxiv.org/abs/2609.36279):.
- C. Benzmüller, "Gödel's and Scott's Variants ... in Lean 4 and TPTP THF", arXiv 2609.26806 v3 (https://arxiv.org/abs/2609.26806): the port the developments build on. It points to 2609.36279 for nominals and the open questions, and has no general-model results.
- C. Benzmüller, "A Comment on Modal Collapse and Ultrafilters", arXiv 2608.07578 (https://arxiv.org/pdf/2608.07578): rigidity of positivity as the driver of collapse. It is background for K1's proviso question and does not address general models.

Our prior-art search (2026-10-23) covered:

- WebFetch: https://arxiv.org/a/benzmuller_c_1 (author listing, 2026 papers), 2026-10-23
- Web pages other than the arXiv listing were seen through search-result summaries only. The Springer and LLP papers were not opened.

The search was run automatically by claude/claude-opus-5-5.

The checkers' full answers, with every query and file they searched, are published with this finding (`prior-art/`).

If you know of related work we missed, please open an issue.

How it was produced

This result was found by AI research agents (the autoresearch harness). It was proved in Lean; the proof and check files are included so anyone can re-run them.

2 Claims

Claim K1

Let T be the $\mathcal{L}$-sentences Ax1GenR at PosGlobal, Ax4, Ax2a, Ax2b of OpenQuestions.Ax1GenRigidVariant2, read off RigidHF.L_R_hf by the translator. Then $P\,G$ (run 4's Run3.K1.Pos Run3.K1.God) holds at every world of every v4 $\mathcal{L}$-general model of T with nonempty individuals.

Status: proved (axioms: propext, Classical.choice, Quot.sound).

Verified. No check scripts.

Evidence.

- proof `checks/K1-TranslateRigid.lean`: Programme.Run6.H.LR.valid_god — pass (`results/K1-4.json`)
- proof `checks/K1-TranslateRigid.lean`: Programme.Run6.H.LR.of_deep — pass (`results/K1-5.json`)
- proof `checks/K1-TranslateRigid.lean`: Programme.Run6.H.LR.of_struct — pass (`results/K1-6.json`)
- reasoning: Structural image check (the theorist's C8). For each sentence of T read off module axiom X with n outer binders the file proves `N.struct<k>: type_of% @X = shV R existsAt P n N.ax<k> ()`, and `N.struct_`φ for the statement, by `rfl` after δ-unfolding the module side; the only rewrites allowed are `(a ↔ b) = ((a → b) ∧ (b → a))` and

(∃ v, R w v ∧ B v) = ¬∀ v, R w v → ¬B v, identities of Lean propositions. shR is the unguarded structural reading, one Lean connective per constructor, so an extra conjunct (the theorist's phiSep mutant, rejected by a fail_if_success test in TranslateV2Mod.lean), a dropped guard or a swapped quantifier fails. N.of_struct re-proves the module statement from the deep certificate through of_valid_full, shV_iff and these equations only, so T and φ are the module's sentences up to definitional unfolding, not just up to full-structure equivalence (the theorist's full_check_blind).

Prior art. Partial prior art.

Claim K3

Let T be the $\mathcal{L}$-sentences Ax1GenTwo, $\exists x.\top$, Ax4, Ax3, Ax2a, Ax2b of OpenQuestions.Ax1GenTwoVariant2, read off the module by the translator. Then unrestricted Ax1Gen holds at every world of every v4 $\mathcal{L}$-general model of T with nonempty individuals (ValidV4).

Status: proved (axioms: propext, Classical.choice, Quot.sound).

Verified. No check scripts.

Evidence.

- proof checks/K3-TranslateTwo.lean: Programme.Run6.H.A1G.valid_ax1gen — pass (results/K3-4.json)
- proof checks/K3-TranslateTwo.lean: Programme.Run6.H.A1G.of_deep — pass (results/K3-3.json)
- proof checks/K3-TranslateTwo.lean: Programme.Run6.H.A1G.of_struct — pass (results/K3-5.json)
- reasoning: Structural image check (the theorist's C8). For each sentence of T read off module axiom X with n outer binders the file proves N.struct<k>: type_of% @X = shV R existsAt P n N.ax<k> (), and N.struct_φ for the statement, by rfl after δ-unfolding the module side; the only rewrites allowed are (a ↔ b) = ((a → b) ∧ (b → a)) and (∃ v, R w v ∧ B v) = ¬∀ v, R w v → ¬B v, identities of Lean propositions. shR is the unguarded structural reading, one Lean connective per constructor, so an extra conjunct (the theorist's phiSep mutant, rejected by a fail_if_success test in TranslateV2Mod.lean), a dropped guard or a swapped quantifier fails. N.of_struct re-proves the module statement from the deep certificate through of_valid_full, shV_iff and these equations only, so T and φ are the module's sentences up to definitional unfolding, not just up to full-structure equivalence (the theorist's full_check_blind).

Prior art. Partial prior art.

Claim K4

Let T be the $\mathcal{L}$-sentences Ax1GenTwo, $\exists x.\top$, Ax4, Ax3, Ax2a, Ax2b of OpenQuestions.Ax1GenTwoVariant3Plain, read off the module by the translator. Then unrestricted Ax1Gen holds at every world of every v4 $\mathcal{L}$-general model of T with nonempty individuals (ValidV4).

Status: proved (axioms: propext, Classical.choice, Quot.sound).

Verified. No check scripts.

Evidence.

- proof `checks/K4-TranslateV3.lean`: Programme.Run6.H.A1G3.valid_ax1gen — pass (`results/K4-4.json`)
- proof `checks/K4-TranslateV3.lean`: Programme.Run6.H.A1G3.of_deep — pass (`results/K4-3.json`)
- proof `checks/K4-TranslateV3.lean`: Programme.Run6.H.A1G3.of_struct — pass (`results/K4-5.json`)
- reasoning: Structural image check (the theorist's C8). For each sentence of T read off module axiom X with n outer binders the file proves `N.struct<k>: type_of% @X = shV R existsAt P n N.ax<k> ()`, and `N.struct_`φ for the statement, by `rfl` after δ-unfolding the module side; the only rewrites allowed are `(a ↔ b) = ((a → b) ∧ (b → a))` and `(∃ v, R w v ∧ B v) = ¬∀ v, R w v → ¬B v`, identities of Lean propositions. `shR` is the unguarded structural reading, one Lean connective per constructor, so an extra conjunct (the theorist's `phiSep` mutant, rejected by a `fail_if_success` test in TranslateV2Mod.lean), a dropped guard or a swapped quantifier fails. `N.of_struct` re-proves the module statement from the deep certificate through `of_valid_full`, `shV_iff` and these equations only, so T and φ are the module's sentences up to definitional unfolding, not just up to full-structure equivalence (the theorist's `full_check_blind`).

Prior art. Partial prior art.

Claim K5

Let T be the $\mathcal{L}$-sentences Ax4, Ax3, Ax1GenTwo, Ax2a, Ax2b, $\exists x.\top$ of OpenQuestions.Ax1GenTwoVariant2, read off its proof of MC_Two_unconditional. Then $\forall p{:}\sigma.\, p \to \Box p$ holds at every world of every v4 $\mathcal{L}$-general model of T with nonempty individuals.

Status: analogy: a resemblance, not an established result.

Verified. No check scripts.

Evidence.

- proof `checks/K5-TranslateV2Mod.lean`: Programme.Run6.H.MCU.valid — pass (`results/K5-3.json`)
- proof `checks/K5-TranslateV2Mod.lean`: Programme.Run6.H.MCU.of_deep — pass (`results/K5-2.json`)
- proof `checks/K5-TranslateV2Mod.lean`: Programme.Run6.H.MCU.of_struct — pass (`results/K5-4.json`)
- reasoning: Structural image check (the theorist's C8). For each sentence of T read off module axiom X with n outer binders the file proves `N.struct<k>: type_of% @X = shV R existsAt P n N.ax<k> ()`, and `N.struct_`φ for the statement, by `rfl` after δ-unfolding the module side; the only rewrites allowed are `(a ↔ b) = ((a → b) ∧ (b → a))` and `(∃ v, R w v ∧ B v) = ¬∀ v, R w v → ¬B v`, identities of Lean propositions. `shR` is the unguarded structural reading, one Lean connective per constructor, so an extra conjunct (the theorist's `phiSep` mutant, rejected by a `fail_if_success` test in TranslateV2Mod.lean), a dropped guard or a swapped quantifier fails. `N.of_struct` re-proves the module statement from the deep certificate through `of_valid_full`, `shV_iff` and these equations only, so T and φ are the module's sentences up to definitional unfolding, not just up to full-structure equivalence (the theorist's `full_check_blind`).

Prior art. Unchecked.

Claim K6

Let T be the $\mathcal{L}$-sentences Ax4, Ax3, Ax1GenTwo, Ax2a, $\exists x.\top$ of OpenQuestions.Ax1GenTwoVariant2, read off its proofs. Then $\Box\exists^E x.G\,x$ and $\neg\Box\neg\exists^E x.G\,x$ hold at every world of every v4 $\mathcal{L}$-general model of T with nonempty individuals.

Status: analogy: a resemblance, not an established result.

Verified. No check scripts.

Evidence.

- proof `checks/K5-TranslateV2Mod.lean`: Programme.Run6.H.Th5U.valid — pass (`results/K6-5.json`)
- proof `checks/K5-TranslateV2Mod.lean`: Programme.Run6.H.Th4.valid — pass (`results/K6-6.json`)

- proof `checks/K5-TranslateV2Mod.lean`: Programme.Run6.H.Th5U.of_deep — pass (`results/K6-3.json`)
- proof `checks/K5-TranslateV2Mod.lean`: Programme.Run6.H.Th4.of_deep — pass (`results/K6-4.json`)
- proof `checks/K5-TranslateV2Mod.lean`: Programme.Run6.H.Th5U.of_struct — pass (`results/K6-7.json`)
- proof `checks/K5-TranslateV2Mod.lean`: Programme.Run6.H.Th4.of_struct — pass (`results/K6-8.json`)
- reasoning: Structural image check (the theorist's C8). For each sentence of T read off module axiom X with n outer binders the file proves `N.struct<k>: type_of% @X = shV R existsAt P n N.ax<k> ()`, and `N.struct_`φ for the statement, by `rfl` after δ-unfolding the module side; the only rewrites allowed are `(a ↔ b) = ((a → b) ∧ (b → a))` and `(∃ v, R w v ∧ B v) = ¬∀ v, R w v → ¬B v`, identities of Lean propositions. `shR` is the unguarded structural reading, one Lean connective per constructor, so an extra conjunct (the theorist's `phiSep` mutant, rejected by a `fail_if_success` test in TranslateV2Mod.lean), a dropped guard or a swapped quantifier fails. `N.of_struct` re-proves the module statement from the deep certificate through `of_valid_full`, `shV_iff` and these equations only, so T and φ are the module's sentences up to definitional unfolding, not just up to full-structure equivalence (the theorist's `full_check_blind`).

Prior art. Unchecked.

Claim K7

Let T be the $\mathcal{L}$-sentences Ax3, Ax2a, Ax2b of OpenQuestions.Ax1GenTwoVariant2, read off its proof of Th3. Then $\neg\Box\neg\exists^E x.G\,x \to \Box\exists^E x.G\,x$ holds at every world of every v4 $\mathcal{L}$-general model of T with nonempty individuals and symmetric R.

Status: analogy: a resemblance, not an established result.

Verified. No check scripts.

Evidence.

- proof `checks/K7-TranslateKB.lean`: Programme.Run6.H.Th3K.validF — pass (`results/K7-1.json`)
- proof `checks/K7-TranslateKB.lean`: Programme.Run6.H.Th3K.of_struct — pass (`results/K7-2.json`)

- reasoning: Frame hypotheses (the theorist's C7). The statement's hypothesis `symm: ∀ x y, x r y → y r x` reifies to the closed frame formula $\forall u\, v.\, R\, u\, v \to R\, v\, u$ (`N.Fs`, checked by `rfl` in the file; `N.Fs_ok: AllRel N.Fs` built constructor by constructor). It is not an $\mathcal{L}$-sentence, so it is not put into T. It stays in the hypothesis list of `N.deep: DW N.Tfull [.w] (N.Fs.map wk) ...`. `sound_v4_wt_frames` (the theorist's `sound_v4_wt_frame` for a list: `toStr` keeps the model's accessibility relation, and a `RelOnly` formula depends on it alone) gives `N.validF: ValidV4Fs N.Fs N.T N.`φ, which quantifies over v4 VM/LGen models with nonempty individuals whose R satisfies every formula of `N.Fs`. A hypothesis that mentions P is not a frame formula; that probe is refused under `#guard_msgs` in the file. `N.of_struct` re-proves the module's statement from `N.validF` through the theorist's `of_validF_full` (for a list), the structural `rfl` checks (C8), and `frame_hold` (the module's `symm` gives `FrameHolds R F`). In that last step only the direction `symm` $\to$ `F` is needed: a weaker reading of the frame condition would make the proved statement stronger, not wrong.

Prior art. Unchecked.

Claim K8

Let T be the $\mathcal{L}$-sentences Ax1Gen, Ax2a, Ax2b, Ax4, $\exists x.\top$ of OpenQuestions.GoedelVariantHOML3inS4oneFile (variant 3), read off its proof of MC_K. Then $\forall p{:}\sigma.\, p \to \Box p$ holds at every world of every v4 $\mathcal{L}$-general model of T with nonempty individuals.

Status: analogy: a resemblance, not an established result.

Verified. No check scripts.

Evidence.

- proof `checks/K8-TranslateS4V3.lean`: Programme.Run6.H.MC3K.valid — pass (`results/K8-1.json`)
- proof `checks/K8-TranslateS4V3.lean`: Programme.Run6.H.MC3K.of_struct — pass (`results/K8-2.json`)
- reasoning: The module's own proofs are translated by `#deep_translate`, which reads the sentences of T off the module axioms the proof uses; lemmas (L, G_ex, Ax2a') are cut in through the theory's prefix (`Good`/`GoodF`). Soundness per output: the kernel checks `N.deep`, and `sound_v4_wt` (run 4) or `sound_v4_wt_frames` turns it into the validity statement. Statement identity: `N.of_struct: type_of% @X` re-proves the module's statement from the certificate through the structural `rfl` checks `N.struct<k>`/`N.struct_`φ (C8). Those checks allow only the rewrites `iff_nf`, `dia_nf`, `exAct_nf`,

which are identities of Lean propositions. Variant 3's Ax4 carries the non-emptiness guard $\varphi \neq \lambda x.\bot$ inside the box, as in the module. No Lean theorem is claimed about the translator itself.

Prior art. Unchecked.

Claim K9

Let T be the $\mathcal{L}$-sentences Ax1Gen, Ax2a, Ax4, $\exists x.\top$ of OpenQuestions.GoedelVariantHOML3inS4oneFile. In every v4 $\mathcal{L}$-general model of T with nonempty individuals, on any frame, $\Box\exists^E x.G\,x$ and $\neg\Box\neg\exists^E x.G\,x$ hold at every world.

Status: analogy: a resemblance, not an established result.

Verified. No check scripts.

Evidence.

- proof `checks/K8-TranslateS4V3.lean`: Programme.Run6.H.Th5V3.valid — pass (`results/K9-1.json`)
- proof `checks/K8-TranslateS4V3.lean`: Programme.Run6.H.Th4S4.validF — pass (`results/K9-2.json`)
- proof `checks/K8-TranslateS4V3.lean`: Programme.Run6.H.Th5V3.of_struct — pass (`results/K9-3.json`)
- proof `checks/K8-TranslateS4V3.lean`: Programme.Run6.H.Th4S4.of_struct — pass (`results/K9-4.json`)
- reasoning: The module's own proofs are translated by `#deep_translate`, which reads the sentences of T off the module axioms the proof uses; lemmas (L, G_ex, Ax2a') are cut in through the theory's prefix (`Good`/`GoodF`). Soundness per output: the kernel checks `N.deep`, and `sound_v4_wt` (run 4) or `sound_v4_wt_frames` turns it into the validity statement. Statement identity: `N.of_struct: type_of% @X` re-proves the module's statement from the certificate through the structural `rfl` checks `N.struct<k>`/`N.struct_`φ (C8). Those checks allow only the rewrites `iff_nf`, `dia_nf`, `exAct_nf`, which are identities of Lean propositions. Variant 3's Ax4 carries the non-emptiness guard $\varphi \neq \lambda x.\bot$ inside the box, as in the module. No Lean theorem is claimed about the translator itself. Th4S4's frame list `Fs` comes from the module axiom `Rrefl`. `ValidV4Fs Fs` quantifies over v4 models whose R satisfies every formula of `Fs`; `Fs_ok: AllRel Fs` is built constructor by constructor.
- proof `checks/K9-TranslateS4V3b.lean`: Programme.Run6.H.Th4B.valid — pass (`results/K9-5.json`)

- proof `checks/K9-TranslateS4V3b.lean`: Programme.Run6.H.Th4B.of_struct — pass (`results/K9-6.json`)
- reasoning: the theorist's ThH19. In variant 3 the inclusion in Ax4 is boxed. At a world with no successor, $G \supset^N \neg G$ is therefore vacuous, and Ax4 with L gives $P\neg G$, which contradicts Ax2a. Th4_box reaches the same contradiction from $\Box\neg\exists^E G$ through G_ex at each successor, with no world-existential step. The translator emits no frame formula. `Th4B.T = Th5V3.T` and `Th4B.`φ `= Th4S4.`φ are checked by rfl.

Prior art. Unchecked.

Claim K10

For the $\mathcal{L}$-sentences T = Ax3, Ax2a, Ax2b and φ = Th3 of K7 (Th3K): φ is not valid in all v4 $\mathcal{L}$-general models of T with nonempty individuals whose R is reflexive and transitive; hence also not without frame condition.

Status: analogy: a resemblance, not an established result.

Verified. No check scripts.

Evidence.

- proof `checks/K10-TranslateKBNec.lean`: Programme.Run6.H.NecSymm.not_validFs_S4 — pass (`results/K10-1.json`)
- proof `checks/K10-TranslateKBNec.lean`: Programme.Run6.H.NecSymm.not_valid — pass (`results/K10-2.json`)
- reasoning: the theorist's two-world S4 model ThH7: R = {00,01,11}, one always-actual entity, positivity P phi w:= phi () 1. Its full structure is L-general (of_validFs_full). Th3K.ax0..2 (Ax3, Ax2b, Ax2a, the very terms emitted for K7) hold there by definitional unfolding of shR (NecSymm.hT uses `exact ThH7.Ax3 w` etc., no rewriting); the closed frame formulas for reflexivity and transitivity hold (NecSymm.frames). So ValidV4Fs \[refl, trans\] Th3K.T Th3K.phi would give Th3 at world 0, contradicting ThH7.Th3_fails; ValidV4 implies ValidV4Fs for any frame list, giving not_valid. World 0 sees world 1 but not conversely, which is exactly where symmetry fails.
- reasoning: Unchanged proof (ThH7 two-world S4 model R = {00,01,11}). The model is neither symmetric nor Euclidean (R00, R01 hold, R10 fails), consistent with K7 and with the Euclidean result; no claim that symmetry is necessary.

Prior art. Unchecked.

Claim K11

Let T be Ax4, Ax3, Ax2a, Ax1GenTwo, $\exists x.\top$ of OpenQuestions.Ax1GenTwoVariant3Plain (plus Ax2b for UniqueEss). Then EssMemI_K_unconditional, UniqueEss1_K_unconditional, UniqueEss2_K_unconditional hold at every world of every v4 $\mathcal{L}$-general model of T with nonempty individuals.

Status: analogy: a resemblance, not an established result.

Verified. No check scripts.

Evidence.

- proof `checks/K11-TranslateV3NE.lean`: Programme.Run6.H.EssMemV3.valid — pass (`results/K11-1.json`)
- proof `checks/K11-TranslateV3NE.lean`: Programme.Run6.H.EssMemV3.of_struct — pass (`results/K11-2.json`)
- proof `checks/K11-TranslateV3NE.lean`: Programme.Run6.H.UE1V3.valid — pass (`results/K11-3.json`)
- proof `checks/K11-TranslateV3NE.lean`: Programme.Run6.H.UE1V3.of_struct — pass (`results/K11-4.json`)
- proof `checks/K11-TranslateV3NE.lean`: Programme.Run6.H.UE2V3.valid — pass (`results/K11-5.json`)
- proof `checks/K11-TranslateV3NE.lean`: Programme.Run6.H.UE2V3.of_struct — pass (`results/K11-6.json`)
- reasoning: Soundness per output: the kernel checks `N.deep: DW...`; `sound_v4_wt` (run 4) gives `N.valid: ValidV4 N.T N.phi`. The not-E rewrite uses only existing DW rules: the forall-E/exists-I instance is the certified L-term M\[E1\], and the switch between M\[E1\] and M\[E0\] is a Leibniz step at type e -\> sigma, which is plain, so LeibOK holds by its first disjunct. Statement identity: `N.of_struct: type_of% @X` reproves the module statement from the certificate through the structural rfl checks `N.struct<k>`/`N.struct_phi` (C8), with only the rewrites iff_nf, dia_nf, exAct_nf. No Lean theorem is claimed about the translator.

Prior art. Unchecked.

Claim K12

Let T be the $\mathcal{L}$-sentences Ax3, Ax2a, Ax2b of OpenQuestions.Ax1GenTwoVariant2. Then $\neg\Box\neg\exists^E x.G\,x \to \Box\exists^E x.G\,x$ holds at every world of every v4 $\mathcal{L}$-general model of T with nonempty individuals and Euclidean R.

Status: analogy: a resemblance, not an established result.

Verified. No check scripts.

Evidence.

- proof `checks/K12-TranslateEucl.lean`: Programme.Run6.H.Th3E.validF — pass (`results/K12-1.json`)
- proof `checks/K12-TranslateEucl.lean`: Programme.Run6.H.Th3E.of_struct — pass (`results/K12-2.json`)
- reasoning: Same mechanism as K7 (C7 frame hypotheses, C8 structural checks). `Th3E.validF: ValidV4Fs Th3E.Fs Th3E.T Th3E.phi` comes from the kernel-checked derivation `Th3E.deep` via `sound_v4_wt_frames`; Fs is the Euclidean frame formula (rfl-checked in the file). `Th3E.of_struct` re-proves the module-level statement (with hypothesis eucl) from validF through of_validFs_full, the structural rfl checks and frame_hold; only the direction eucl -\> F is used. The source proof is the theorist's ThH10.Th3_eucl, copied verbatim; no Lean theorem is claimed about the translator.

Prior art. Unchecked.

Claim K13

Let T be Ax1Gen, Ax2a, Ax3, Ax4, $\exists x.\top$ of OpenQuestions.GoedelVariantHOML3inS4oneFile (plus Ax2b for UniqueEss). Then EssMemI_K, UniqueEss1_K, UniqueEss2_K, Ess_iff_Ess2_K hold at every world of every v4 $\mathcal{L}$-general model of T with nonempty individuals.

Status: analogy: a resemblance, not an established result.

Verified. No check scripts.

Evidence.

- proof `checks/K13-TranslateS4NE.lean`: Programme.Run6.H.EssMem3S.valid — pass (`results/K13-1.json`)
- proof `checks/K13-TranslateS4NE.lean`: Programme.Run6.H.EssMem3S.of_struct — pass (`results/K13-2.json`)
- proof `checks/K13-TranslateS4NE.lean`: Programme.Run6.H.UE1S.valid — pass (`results/K13-3.json`)
- proof `checks/K13-TranslateS4NE.lean`: Programme.Run6.H.UE1S.of_struct — pass (`results/K13-4.json`)
- proof `checks/K13-TranslateS4NE.lean`: Programme.Run6.H.UE2S.valid — pass (`results/K13-5.json`)

- proof `checks/K13-TranslateS4NE.lean`: Programme.Run6.H.UE2S.of_struct — pass (`results/K13-6.json`)
- proof `checks/K13-TranslateS4NE.lean`: Programme.Run6.H.EssIffS.valid — pass (`results/K13-7.json`)
- proof `checks/K13-TranslateS4NE.lean`: Programme.Run6.H.EssIffS.of_struct — pass (`results/K13-8.json`)
- reasoning: Same mechanism as K11 (not-E rewrite: forall-E/exists-I at the certified L-term M\[E1\], E1 = lambda z. exists^E y. y = z, switched to M\[E0\] by a Leibniz step at the plain type e -\> sigma, so LeibOK holds by its first disjunct). The kernel checks N.deep; sound_v4_wt gives N.valid: ValidV4 N.T N.phi. No frame formula is emitted: the module proofs use the lemma Reach_K, not Rrefl/Rtrans, so the result holds on arbitrary frames. N.of_struct re-proves the module statement from N.valid through the structural rfl checks (C8; rewrites iff_nf, dia_nf, exAct_nf only). The module's own proofs are translated unchanged. No Lean theorem is claimed about the translator.

Prior art. Unchecked.

Claim K14

Let T = Ax3, Ax2b, Ax2a of OpenQuestions.Ax1GenTwoVariant2 (K7's Th3K) and φ = Th3. For every list F_s of frame formulas true on F4 = $\{01, 03, 31, 12, 21\}$, φ is not valid in all v4 $\mathcal{L}$-general models of T with nonempty individuals whose R satisfies F_s.

Status: analogy: a resemblance, not an established result.

Verified. No check scripts.

Evidence.

- proof `checks/K14-TranslateKBF4.lean`: Programme.Run6.H.F4.not_validFs_F4 — pass (`results/K14-1.json`)
- proof `checks/K14-TranslateKBF4.lean`: Programme.Run6.H.F4.F4_Qt — pass (`results/K14-2.json`)
- reasoning: Pull-back along a surjective bounded morphism (forth/back) keeps carriers and evaluation; only the box clause uses forth/back, pr_inj uses surjectivity (the theorist's ThH13). The refuting model on F4 is the pull-back of the full model on F3, which is L-general by ofStr_lgen. F4_Qt: no world of F4 sees a quasi-terminal world together with a different world.

Prior art. Unchecked.

Claim K15

For T = Ax3, Ax2b, Ax2a, φ = Th3 of Ax1GenTwoVariant2: on any frame (W, R) with B R-closed, uniform (all members have successors or none), $b \in B$, $u \notin B$, Rwb, Rwu, some v4 $\mathcal{L}$-general model of T on (W, R), nonempty individuals, refutes φ.

Status: analogy: a resemblance, not an established result.

Verified. No check scripts.

Evidence.

- proof `checks/K15-TranslateKBQL.lean`: Programme.Run6.H.QL.not_validOn — pass (`results/K15-1.json`)
- proof `checks/K15-TranslateKBQL.lean`: Programme.Run6.H.QL.not_validFs — pass (`results/K15-2.json`)
- reasoning: col B is a surjective bounded morphism from (W,R) onto RC R B when B is nonempty, R-closed and uniform (col_back). Pulling an L-general model back along a surjective bounded morphism keeps it L-general with the same evaluation (pullL, the theorist's ThH13), and its accessibility is R by definition. The full model on RC R B with P phi:= phi () (inr ()) satisfies Ax3 because inr () is quasi-terminal (pt_qt), satisfies Ax2a/Ax2b trivially, and refutes Th3 at col w.

Prior art. Unchecked.

Claim K16

Let T = Ax3, Ax2b, Ax2a of OpenQuestions.Ax1GenTwoVariant2. On any frame (W, R) with Q_L (no world outside an R-closed B sees a member and a non-member of B), $\Diamond\exists^E G \to \Box\exists^E G$ holds everywhere in every v4 $\mathcal{L}$-general model of T with nonempty individuals.

Status: analogy: a resemblance, not an established result.

Verified. No check scripts.

Evidence.

- proof `checks/K16-TranslateKBQLi.lean`: Programme.Run6.H.QLi.validOn_QL — pass (`results/K16-1.json`)
- proof `checks/K16-TranslateKBQLi.lean`: Programme.Run6.H.Th2X.valid — pass (`results/K16-2.json`)
- proof `checks/K16-TranslateKBQLi.lean`: Programme.Run6.H.QLi.validOn_symm — pass (`results/K16-3.json`)

- proof `checks/K16-TranslateKBQLi.lean`: Programme.Run6.H.QLi.validOn_eucl — pass (`results/K16-4.json`)
- reasoning: Th2X.valid gives $\psi_G \to \Box\psi_G$ at every world of every L-general model of Th3K.T (T_eq, th2_eq by rfl). Then the extension B of ψ_G is R-closed. If $w \models \Diamond\psi_G$, some $b \in B$ with Rwb. If $w \in B$, closure gives $\Box\psi_G$; otherwise Q_L puts every successor u of w in B. Only the LGen clauses imp, neg, box are used (th3_of_th2). th3_eq identifies the result with Th3K.φ by rfl.

Prior art. Unchecked.

Claim K17

Let T = Ax3, Ax2b, Ax2a of OpenQuestions.Ax1GenTwoVariant2. On any frame (W, R), Th3 holds everywhere in every v4 $\mathcal{L}$-general model of T with nonempty individuals iff no world outside an R-closed set B of all-dead or all-serial worlds sees a member and a non-member of B.

Status: analogy: a resemblance, not an established result.

Verified. No check scripts.

Evidence.

- proof `checks/K17-TranslateKBQU.lean`: Programme.Run6.H.QU.validOn_iff_QU — pass (`results/K17-1.json`)
- proof `checks/K17-TranslateKBQU.lean`: Programme.Run6.H.QU.F5_valid — pass (`results/K17-2.json`)
- proof `checks/K17-TranslateKBQU.lean`: Programme.Run6.H.QU.F5_not_QL — pass (`results/K17-3.json`)
- proof `checks/K17-TranslateKBQU.lean`: Programme.Run6.H.DPos.valid — pass (`results/K17-4.json`)
- proof `checks/K17-TranslateKBQU.lean`: Programme.Run6.H.DRigN.valid — pass (`results/K17-5.json`)
- reasoning: If: from Th2X.valid, DPos.valid, DRig.valid, DRigN.valid (kernel-checked derivations, sound_v4_wt) the four sentences $\psi \to \Box\psi$, $\psi \to (\Box\bot \leftrightarrow \rho)$, $\rho \to \Box\rho$, $\neg\rho \to \Box\neg\rho$ hold everywhere ($\psi = \exists^E G$, $\rho = P(\lambda z.\Box\bot)$, all matched by rfl). Then $B = \{v \mid \psi v \wedge (\rho v \leftrightarrow \rho b)\}$ is R-closed and uniform, and Q_U gives $\Box\psi$ (th3_of_unif). Only if: a closed uniform violating B gives K15's refuting model (collapse plus pull-back).
- proof `checks/K17-TranslateKBQC.lean`: Programme.Run6.H.QC.validOn_iff_QC — pass (`results/K17-6.json`)

- proof `checks/K17-TranslateKBQC.lean`: Programme.Run6.H.QC.QU_iff_QC — pass (`results/K17-7.json`)
- proof `checks/K17-TranslateKBQC.lean`: Programme.Run6.H.QC.dead_only_of_validOn — pass (`results/K17-8.json`)
- reasoning: the theorist's ThH18 (verbatim). For $Q_U \to Q_C$, take $B = R^*(b)$, which is closed. If $w \in R^*(b)$, closure gives $u \in R^*(b)$; otherwise Q_U applies. For $Q_C \to Q_U$, a closed $B \ni b$ contains $R^*(b)$, so uniformity of B passes to the cone, and then $u \in R^*(b) \subseteq B$.

Prior art. Unchecked.

Provenance. Run `20261023-0556`, claims K1, K3, K4, K5, K6, K7, K8, K9, K10, K11, K12, K13, K14, K15, K16, K17, run-folder commit `ade4f20858f2a3b683193e13de0d89273d57c4c9`, published 2026-10-24.

3 How to verify

Each finding has its check scripts in `claims/ontological-proof-translation/checks/` of the findings repository. Run each with `python -I <script>`; it passes if it exits with status 0. The checks are Python scripts and need the packages of the research image (the image's `requirements.txt`); the Lean checks also need `lean`.

4 Revisions

- 2026-10-24 — readable prior-art section

References

[1] Mihaly Barasz, Paul Christiano, Benja Fallenstein, Marcello Herreshoff, Patrick LaVictoire, and Eliezer Yudkowsky. Robust cooperation in the prisoner's dilemma: Program equilibrium via provability logic, 2014.

[2] Paul Beame and Søren Riis. More on the relative strength of counting principles. In *Proof Complexity and Feasible Arithmetics*, volume 39 of *DIMACS Series in Discrete Mathematics and Theoretical Computer Science*, pages 13–35. American Mathematical Society, 1998.

[3] Christoph Benzmüller. A (simplified) supreme being necessarily exists, says the computer: Computationally explored variants of gödel's ontological argument. In *Proceedings of KR 2020*, 2020.

[4] Christoph Benzmüller. A comment on modal collapse and ultrafilters in gödel's ontological argument. arXiv:2608.07578, 2026.

[5] Christoph Benzmüller. Gödel's and scott's variants of the ontological argument in lean 4 and tptp thf. arXiv:2609.26806, 2026.

[6] Christoph Benzmüller. Proofs without nominals: Gödel's ontological argument, its shallow embedding, and the open questions of the monatshefte notes. arXiv:2609.36279v4, 2026.

[7] Christoph Benzmüller and David Fuenmayor. Computer-supported analysis of positive properties, ultrafilters and modal collapse in variants of gödel's ontological argument. arXiv:1910.08955, 2019.

[8] Christoph Benzmüller and Lawrence C. Paulson. Quantified multimodal logics in simple type theory. *Logica Universalis*, 7(1), 2013.

[9] Christoph Benzmüller and Dana S. Scott. Notes on gödel's and scott's variants of the ontological argument. *Monatshefte für Mathematik*, 2025.

[10] Christoph Benzmüller, Leon Weber, and Bruno Woltzenlogel Paleo. Computer-assisted analysis of the anderson–hájek ontological controversy. *Logica Universalis*, 2017.

[11] Patrick Blackburn, Maarten de Rijke, and Yde Venema. *Modal Logic*. Cambridge University Press, 2001.

[12] George Boolos. *The Logic of Provability*. Cambridge University Press, 1993.

[13] Andrew Critch. A parametric, resource-bounded generalization of löb's theorem, and a robust cooperation criterion for open-source game theory. *Journal of Symbolic Logic*, 2019.

[14] Andrew Critch, Michael Dennis, and Stuart Russell. Cooperative and uncooperative institution designs: Surprises and problems in open-source game theory. arXiv:2208.07006, 2022.

[15] Kit Fine. Propositional quantifiers in modal logic. *Theoria*, 36(3), 1970.

[16] Melvin Fitting. *Types, Tableaus, and Gödel's God*. Kluwer, 2002.

[17] Murdoch J. Gabbay and Dominic P. Mulligan. Nominal Henkin semantics: simply-typed lambda-calculus models in nominal sets. arXiv:1111.0089, 2011.

[18] Gram_Stone. Modal chicken. LessWrong, 2015.

[19] Donald Hobson. Yet more modal combat. LessWrong / AI Alignment Forum, 2021.

[20] Wesley H. Holliday. A note on algebraic semantics for S5 with propositional quantifiers. *Notre Dame Journal of Formal Logic*, 60(2), 2019.

[21] Patrick LaVictoire, Benja Fallenstein, Eliezer Yudkowsky, Mihaly Barasz, Paul Christiano, and Marcello Herreshoff. Program equilibrium in the prisoner's dilemma via löb's theorem. In *AAAI Workshop on Multiagent Interaction without Prior Coordination*, 2014.

[22] Alex Mennen. Modal combat for games other than the prisoner's dilemma. LessWrong / AI Alignment Forum, 2017.

[23] Mladen Mika and Jakob Nordström. A generalized method for proving polynomial calculus degree lower bounds. In *30th Conference on Computational Complexity (CCC 2015)*, volume 33 of *LIPIcs*, pages 467–487, 2015.

[24] Reinhard Muskens. Intensional models for the theory of types. *Journal of Symbolic Logic*, 72(1), 2007.

[25] Caspar Oesterheld. A note on the compatibility of different robust program equilibria of the prisoner's dilemma, 2022.

[26] Alexander A. Razborov. Proof complexity of pigeonhole principles. In *Developments in Language Theory (DLT 2001)*, volume 2295 of *Lecture Notes in Computer Science*, pages 100–116. Springer, 2002.

[27] Søren Riis. *Independence in Bounded Arithmetic*. PhD thesis, University of Oxford, 1993.

[28] Søren Riis and Meera Sitharam. Non-constant degree lower bounds imply linear degree lower bounds. Technical Report TR97-048, Electronic Colloquium on Computational Complexity, 1997.

[29] Robert M. Solovay. Provability interpretations of modal logic. *Israel Journal of Mathematics*, 25:287–304, 1976.

[30] Aleksandar Todorov, Jesse ten Napel, and Alexander Müller. Parametric open source games. arXiv:2606.27068, 2026.

[31] transhumanist_atom_understander. Is there only one FairBot? LessWrong, with a comment by Zac Hatfield-Dodds, 2026.